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LLM-Assisted Financial Statement Analysis Tool

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1. Abstract

The project is a web-based tool for analyzing financial statements using automated extraction, financial calculations, and an LLM-based explanation layer.

Users can upload financial statements in PDF or Excel format.

The system extracts financial data, calculates important ratios, identifies trends, and generates meaningful insights.

The LLM explains validated analytical results in simple, understandable language.

A dashboard presents key results visually and a report provides detailed analysis.

2. Literature Survey

FinQA (Chen et al., EMNLP 2021) introduced a financial question-answering benchmark focused on numerical reasoning over financial reports.

FinQA combines structured tables and surrounding text and includes annotated reasoning programs for explainability.

Financial-report QA research shows that finance requires document retrieval, table understanding, and multi-step numerical reasoning.

Recent research explores language models for numerical reasoning over financial reports.

Research gap addressed by this project: one workflow combining extraction, financial analysis, trends, LLM explanation, dashboard visualization, and reporting.

3. Problem Statement

Financial statements contain large amounts of structured and unstructured information.

Manual extraction and calculation of financial metrics can be time-consuming.

Users may find ratios and year-to-year changes difficult to interpret.

Traditional tools often separate calculation, visualization, and explanation.

A unified system is needed to automate analysis and explain results in simple language.

4. Existing System

Manual review of financial statements and tables.

Spreadsheet-based calculations for ratios and comparisons.

Separate tools may be needed for extraction, analysis, visualization, and reporting.

Interpretation often depends on the user's financial knowledge.

Limitations include time-consuming work, fragmented tools, and limited natural-language explanation.

5. Proposed System

Workflow: Upload → Extract → Clean & Validate → Analyze → Trends → Insights → LLM Explanation → Dashboard → Report.

Supports PDF and Excel financial statements.

Uses deterministic calculations for financial ratios and trends.

Uses an LLM mainly to explain validated results in natural language.

Provides visual summaries and a detailed report from the same analysis pipeline.

Financial calculations are kept separate from LLM interpretation to reduce invented numerical results.

6. Hardware Requirements

Processor: Intel Core i5 / AMD Ryzen 5 or equivalent.

RAM: Minimum 8 GB; 16 GB recommended for development.

Storage: At least 10 GB free space for the project and test documents.

Display: 1366×768 or higher.

Internet connection: required when using a cloud-based LLM/API.

No dedicated GPU is required for the basic web application with an external LLM API.

7. Software Requirements

Operating System: Windows / Linux / macOS.

Programming Language: Python.

Backend: Flask.

Data processing: Pandas and NumPy.

Excel processing: OpenPyXL.

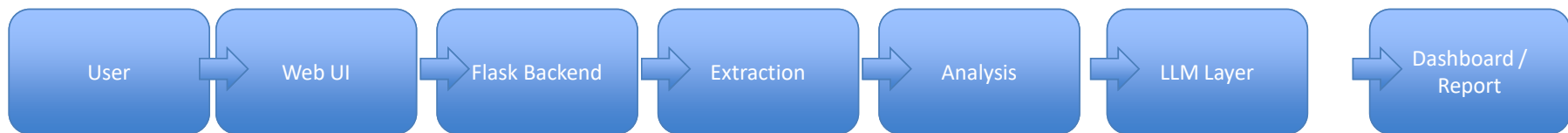
PDF processing: PyMuPDF.

Frontend: HTML, CSS, JavaScript.

LLM/API layer: replaceable LLM service or compatible API.

Development: VS Code or equivalent IDE; modern web browser.

8. Software Architecture



Input layer accepts financial statements.

Backend coordinates validation, extraction, analysis, and output generation.

Analysis engine performs numerical calculations and trend detection.

LLM layer explains validated results.

Presentation layer displays the dashboard and generates the report.

9. Modules

1. File Upload & Validation — accepts PDF/Excel files and validates input.
2. Data Extraction — extracts tables and financial values.
3. Data Cleaning & Validation — standardizes values and checks missing/inconsistent data.
4. Financial Analysis — calculates liquidity, profitability, and leverage ratios.
5. Trend & Insight — identifies changes, strengths, weaknesses, and unusual movements.
6. LLM Explanation — converts validated analysis into simple language.
7. Dashboard — presents KPIs, ratios, charts, trends, and insights.
8. Report Generation — creates a detailed financial analysis report.

10. UML / Use-Case Diagram



Project Flow & Expected Outcome

Financial Statement → Structured Data → Ratios & Trends → Insights → LLM Explanation.

Dashboard provides a quick visual understanding of financial performance.

Report provides a detailed and understandable analysis.

The system reduces manual effort while keeping numerical calculations transparent and verifiable.

Goal: make financial statement analysis easier for users with limited financial expertise.